# Adnan Harun Dogan

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## Education

**Master of Science in Computer Engineering** — *Sep 2022 – Jan 2025*
*Middle East Technical University (METU)* — *Ankara, Turkey*

- **Thesis:** Incorporating Piecewise Linear Functions with Constant Regions in Backpropagation
- **Advisor:** Prof. Dr. Sinan Kalkan, Assoc. Prof. Dr. Emre Akbas

**Bachelor of Science in Computer Engineering** — *Sep 2018 – Jul 2022*
*Middle East Technical University (METU)* — *Ankara, Turkey*

## Experience

**Graduate Intern** (Supervisor: Prof. Dr. Christian Holz) — *Mar 2025 – Oct 2025*
*Sensing, Interaction, & Perception Laboratory (SIPLab), ETH Zürich* — *Zürich, Switzerland*

- Developed learned frequency-weighted Kalman filters that denoise the innovation spectrum under spectral supervision
- Built EEG multitaper-spectrum deep models (ConvLSTM) for real-time VR cybersickness detection, +30% over baselines

**Research Assistant** (Supervisor: Prof. Dr. Sinan Kalkan, Assoc. Prof. Dr. Emre Akbas) — *Feb 2023 – Present*
*Image Processing and Pattern Recognition Laboratory (ImageLab), METU* — *Ankara, Turkey*

- Contributed to Bucketed Ranking Loss, speeding up ranking-based training of DETR/transformer detectors by up to 6x
- Developed optimal-transport (Sinkhorn-Knopp) and ranking-based loss formulations for end-to-end object detection

**Undergraduate Intern** (Supervisor: Assoc. Prof. Dr. Erol Sahin, Assoc. Prof. Dr. Bugra Koku) — *Oct 2021 – Feb 2022*
*Center for Artificial Intelligence and Robotics (ROMER), METU* — *Ankara, Turkey*

- Built a self-driving stack for a physical RC car, and trained RL agents for autonomous navigation task in ROS/Gym/PyTorch
- Simulated a Hector-quadrotor swarm in ROS/Gazebo, approximating numerical solvers for multi-agent localization

**Undergraduate Intern** (Supervisor: Prof. Dr. Mohammad Abdullah Al-Faruque) — *Jun 2021 – Sep 2021*
*Cyber-Physical Systems Laboratory, University of California, Irvine* — *Irvine, California, USA*

- Built multi-modal deep models for beat-by-beat cardiac and sleep abnormality detection from physiological signals
- Fused time-series and spectral features for arrhythmia classification, competitive in international challenges

## Publications

**Dogan, A. H.**, Demirel, B. U., Holz, C. (2026). *Deep Kalman State Estimation for Drift-Calibrated Full-Body Pose Tracking from Sparse Inertial Sensors.* [Under Review]

**Dogan, A. H.**, Demirel, B. U., Holz, C. (2026). *Magnetometer-Free State Estimation for Sparse Inertial Human Pose Tracking via Motion-Based Yaw Inference.* [Under Review]

**Dogan, A. H.**, Deniz, M., Alemdar, H., Baran, O. U. (2026). *CoFINN: Conservation Flux Informed Neural Networks for Physics Problems Governed by Conservation Laws.* [Under Review]

**Dogan, A. H.**, Demirel, B. U., Holz, C. (2026). *Frequency-Weighted Neural Kalman Filters.* **ICRA 2026**. [Code]

Demirel, B. U., **Dogan, A. H.**, Rossie, J., Möbus M., Holz, C. (2025). *Beyond subjectivity: Continuous cybersickness detection using eeg-based multitaper spectrum estimation.* **IEEE/TVCG 2025**. [DOI | Code]

Yavuz, F., Cam B.C., **Dogan, A. H.**, Oksuz, K., Kalkan, S., Akbas, E. (2024). *Bucketed Ranking Loss for Efficient Ranking-based Training of Object Detectors.* **ECCV 2024**. [DOI | PDF | Code]

Amin M. A., **Dogan, A. H.**, Kuru E. S., Sever Y., Angin P. (Apr 2024). *Misuse Detection and Response for Orchestrated Microservices Based Software.* AINA 2024. [PDF | Data]

Karagoz, P., Cekinel, R. F., **Dogan, A. H.**, Oktay, B., Ozturk, A. U., Tonay, S. T., Tunel, B. M. (2024). *Enhancing Underground Built Heritage Analysis with Text Mining: A Case Study on Cappadocia.* Book Chapter 2024. [PDF]

Sever Y., **Dogan, A. H.** (Aug 2023). *A Kubernetes dataset for misuse detection.* ITU Journal 2023. [DOI | PDF | Code]

Sever, Y., Ekinci, G., **Dogan, A. H.**, Alparslan, B., Gurbuz, A. S., Jabrayilov, V., Angin, P. (Sep 2022). *An Empirical Analysis of IDS Approaches in Container Security.* IEEE/SRMC 2022. [Best Paper Award] [DOI]

Demirel, B. U., **Dogan, A. H.**, Al-Faruque, M. A. (2021). *Two Might Do: A Beat-by-Beat Classification of Cardiac Abnormalities using Deep Learning and Domain-Specific Features.* CinC 2021. [DOI | PDF | Code]

Buyukbas, E. B., **Dogan, A. H.**, Ozturk, A. U., Karagoz, P. (Sep 2021). *Explainability in Irony Detection.* DaWaK 2021. [PDF]

**Dogan, A. H.**, Dogan, A. (Jun 2021). *An Assembled Deep Learning Approach for Flow Field Prediction.* [PDF]

## Scores, Honors, & Awards

**TOEFL iBT: 109/120 (R29 L29 S25 W26)**, ETS — *Oct 2022*
**University Entrance Exam: ranked 1466th of 2.27M candidates**, OSYM, Turkey — *Aug 2017*

## Technical Skills & Research Interests

**Languages & Tools**: Python/PyTorch, C++, ROS2, Rust
**DevOps & Containers**: GitHub, Docker, Singularity, Apptainer
**Differentiable Optimization**: Combinatorial Optimization, Optimal Transport, Deep State Space Models
**Deep Learning**: Computer Vision, Self-Supervised Learning, Probabilistic Deep Learning, Deep Reinforcement Learning